Manas Twin

A digital twin that warns of a depressive episode about a week before it begins

For Indian college students and young professionals, and the counsellors who support them



Happiest Health · Reimagining and Reforming Healthcare in India Summit 2026, Bengaluru
Digital Twin Proof-of-Concept Challenge · Team Kabil · Shreyansh Joshi (Team Lead)
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India's mental-health care starts too late: the treatment gap exceeds 60% for almost every disorder

~15%

of Indian adults need treatment for one or more mental health conditions

>60%

treatment gap for every mental disorder except epilepsy (range 28–83%)

Why care starts late

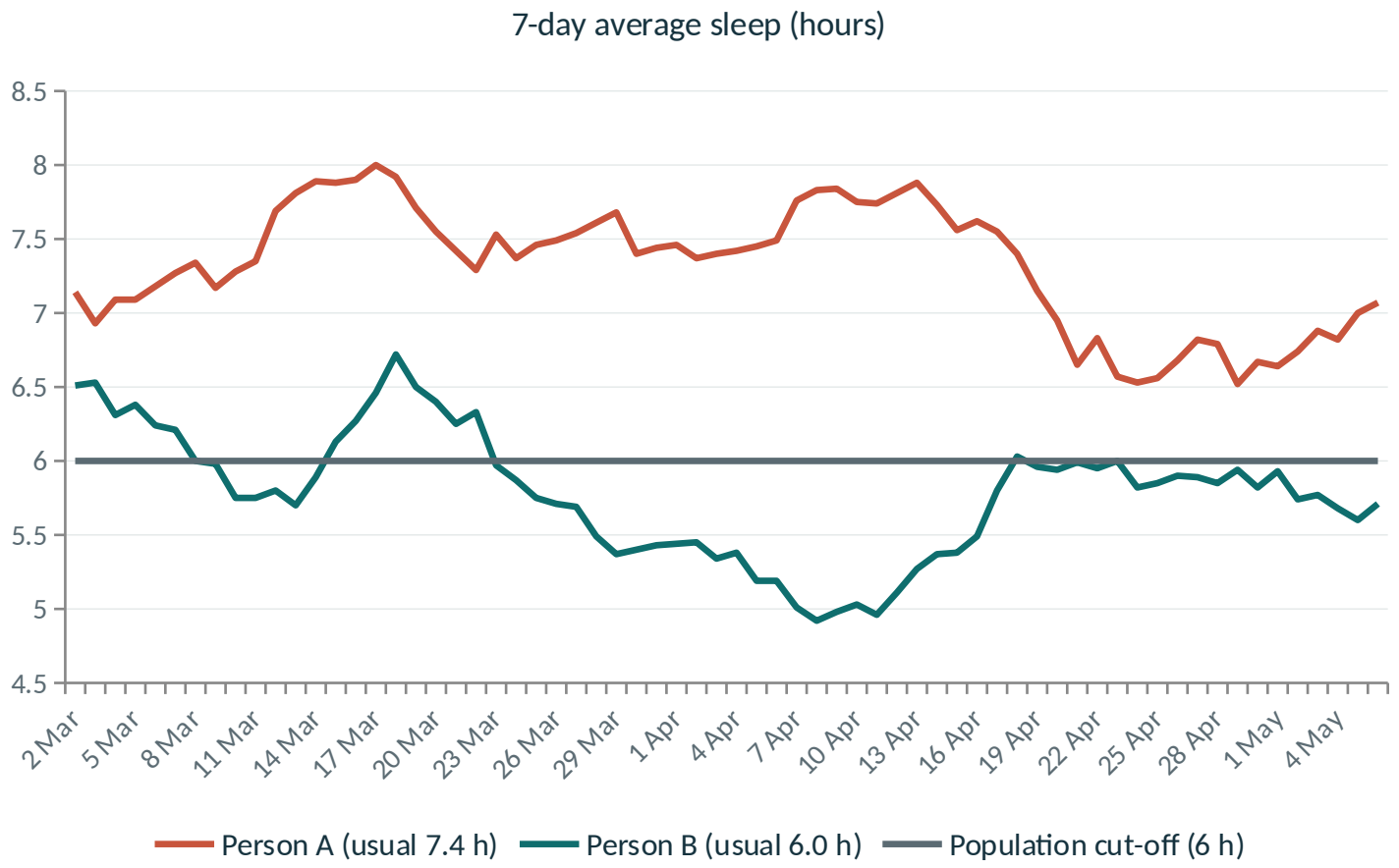
Symptoms build silently for weeks
Sleep, energy, activity and social contact change before anyone names it

Help-seeking is slow
Stigma and few counsellors mean diagnosis often comes well into an episode

Nobody watches daily signals
Wearables and phones already record the build-up, but no one reads it for the person

Verified fact: NIMHANS, National Mental Health Survey of India 2015–16 (34,802 adults, 12 states). Right panel: problem framing.

Everyone's 'normal' is different, so one population cut-off flags the wrong person



Person A: above the cut-off, heading into an episode
Sleep fell almost an hour below their own normal. Twin: Early warning, 6.43% risk (7× typical).

Person B: below the cut-off, but stable
A natural short sleeper. Twin risk never above 0.5%.

Synthetic held-out test patients MT0441 (A) and MT0178 (B), 2 Mar – 6 May. The 6 h cut-off is illustrative.

Built for students and young professionals, and the counsellors who support them

Student / employee
Daily 'you vs your normal' mirror, gentle nudges, what-if view. Joins with explicit consent.

Counsellor / EAP / doctor
Risk-ranked caseload with reasons and PHQ-9 trend, so they can reach out before a crisis.

Institution (aggregate)
Anonymised view of stress hotspots like exam weeks, to plan counselling capacity.

How an alert becomes care

1. Wearable + phone data arrive daily
2. Twin sees drift from personal normal
3. Early warning with plain-language reasons
4. Counsellor check-in and repeat PHQ-9
5. Care starts in week zero, not week four

The twin does four things for each person: mirror, predict, explain, simulate

Mirror

Compares your last 3 and 7 days with your own 45-day normal across 8 signals, plus a single Drift Index

Predict

Probability that a depressive episode begins within the next 7 days, updated every day

Explain

Top reasons in plain language: "steps 18% below usual", "54 min less sleep" (SHAP)

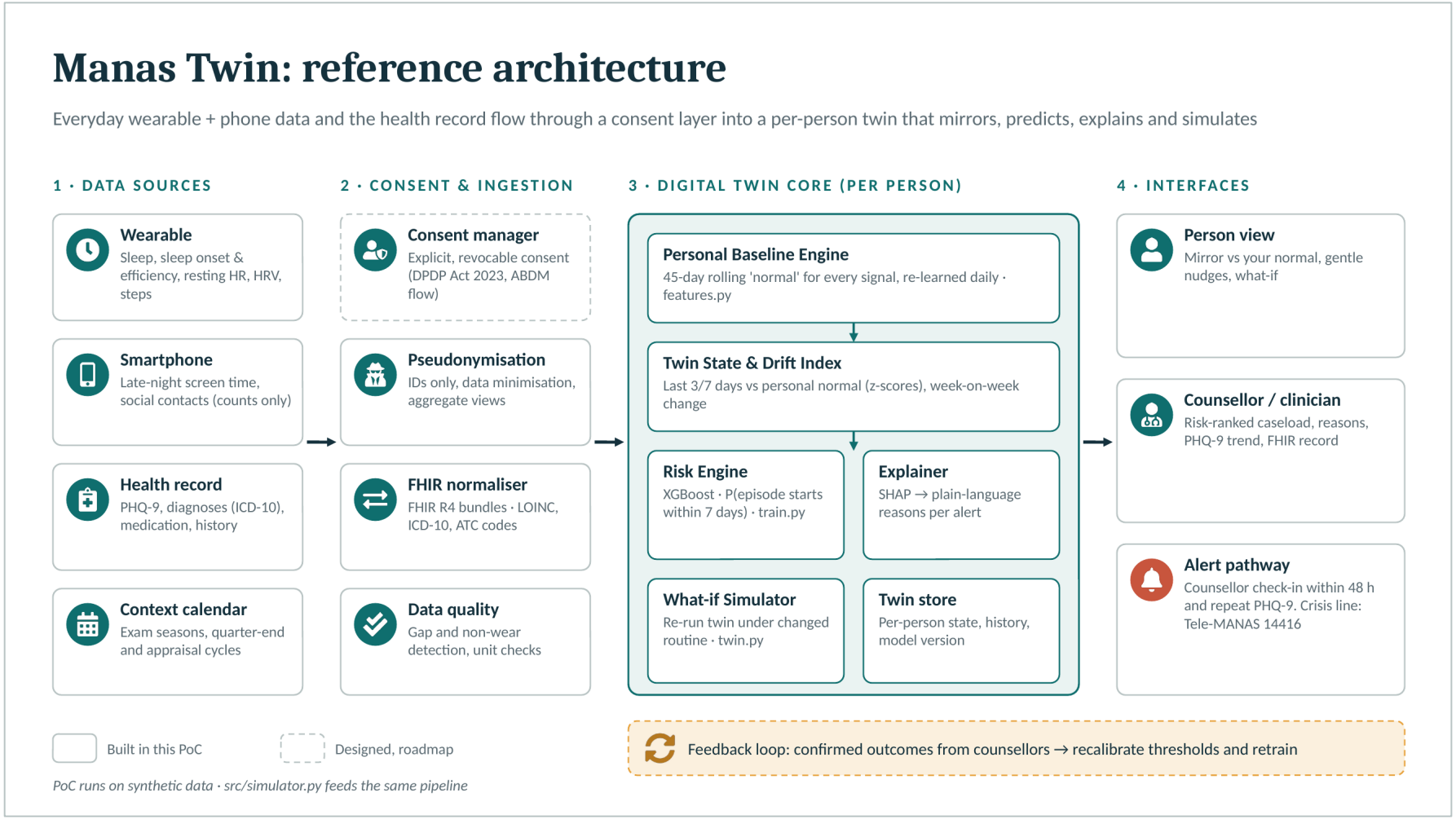
Simulate

What-if: change sleep, bedtime, steps, screen time or social contact and re-run the twin

Signals: Sleep duration, onset, efficiency · Resting HR, HRV · Daily steps · Late-night screen time, social contacts

+ health record: PHQ-9, prior episodes, family history, medication, comorbidities · exam / work-crunch calendar

A consent-first pipeline turns everyday data into an explainable 7-day risk



Full-size diagram: docs/Manas_Twin_Architecture.pdf

No public dataset pairs wearables with records, so we simulated 600 people, then checked the features on real actigraphy

600

synthetic people, 12 Indian cities

144,000

person-days (240 days each)

175

depressive episodes in 148 people

8 + EHR

daily signals plus health record

How the simulator works

Drivers
Vulnerability, prior episodes, family history, exam / work-crunch calendar, random life events

↓

Hidden depressive load
Builds up gradually; an episode starts when it crosses a threshold and lasts 14+ days

↓

What the twin sees
Signals drifting from each person's own baseline, plus PHQ-9, diagnoses and medication

Synthetic data generated by src/simulator.py. Signal directions follow digital-phenotyping literature; magnitudes are modelling assumptions.

Personal-baseline features + XGBoost + SHAP: accurate enough to screen, transparent enough to trust

50 twin features

Per signal: 7-day mean, deviation from personal normal (3- and 7-day), week-on-week change · Drift Index · non-wear days · EHR context

→

XGBoost risk engine

350 trees, depth 4 · handles missing days natively · P(episode starts within 7 days)

→

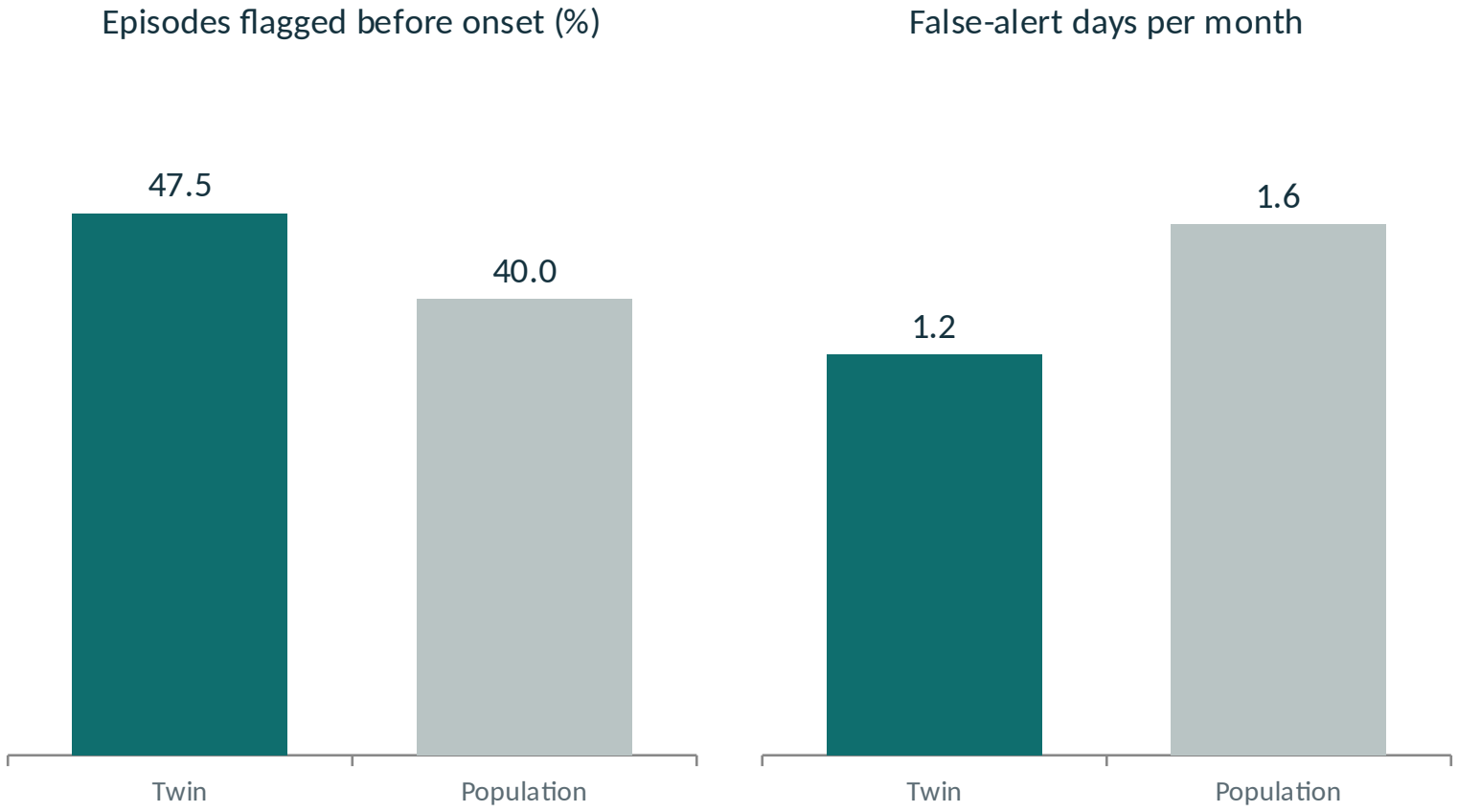
SHAP + alert bands

Reasons for every score · Early warning at ≥3% risk (≈3× base rate), Watch at half that

Fair test

Split by person: 420 train / 180 unseen test. Threshold fixed on validation people at 95% specificity. Benchmarked against an identical model without personal-baseline features (the 'population model').

Personalisation catches more episodes in advance with about 25% fewer false alerts



0.85

AUROC on unseen synthetic people (population model 0.81)

6 days

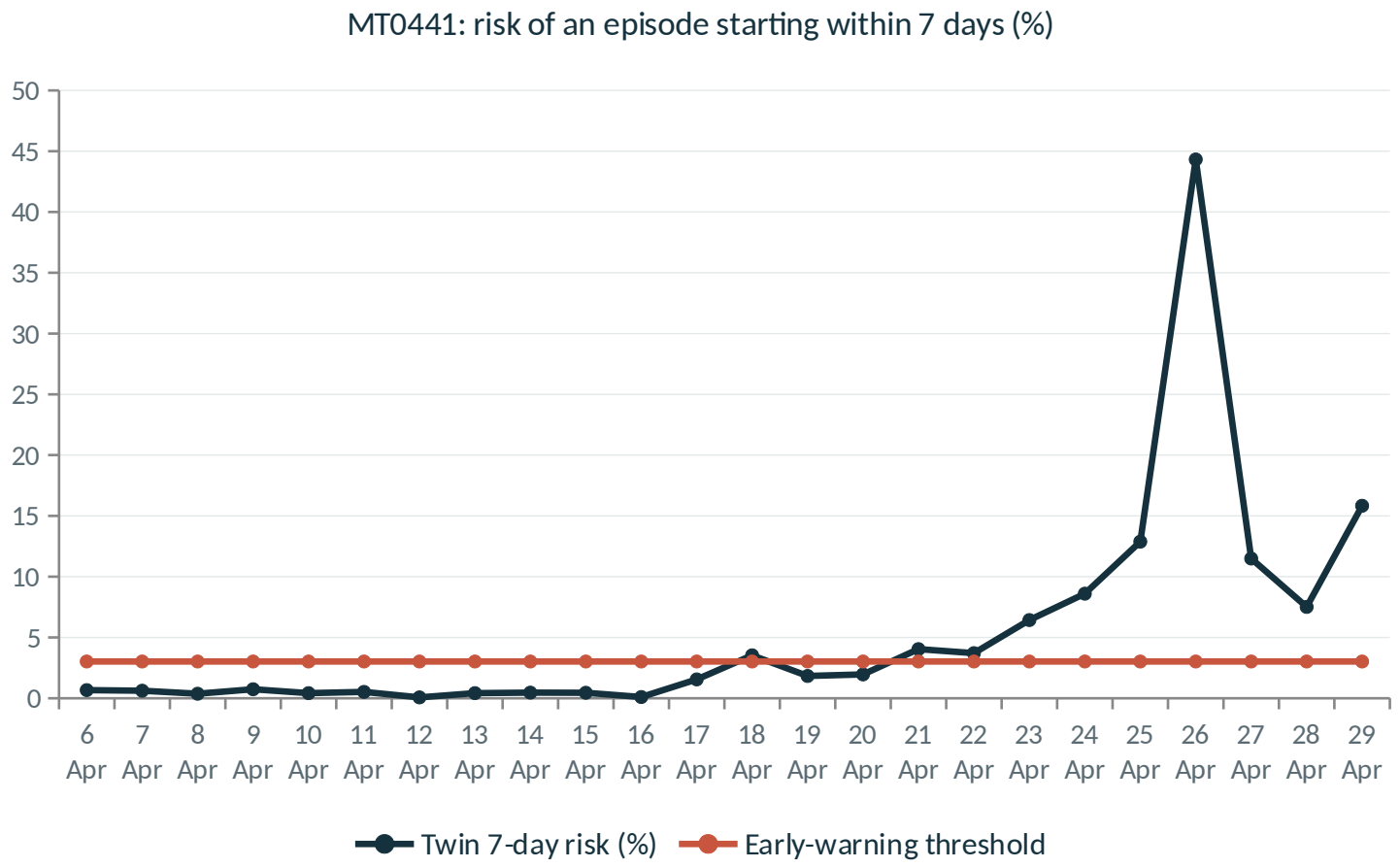
median warning before the episode begins

0.82

AUROC on REAL wrist-actigraphy, depressed vs control (55 people, 95% CI 0.71–0.92)

Charts: synthetic held-out test set (180 people, 40 evaluable episodes). Real-data check: public DEPRESJON dataset (Garcia-Ceja et al., 2018), leave-one-person-out; it tests the features, not the 7-day early warning.

For one student, the twin raised the alarm 7 days before onset; usual care diagnosed 19 days after



21 Apr Twin: Early warning
Sleep and steps falling below his own normal during exams

28 Apr Episode begins
Ground truth from the simulator, hidden from the twin

17 May Usual care: diagnosis
PHQ-9 of 16 at clinic, escitalopram started

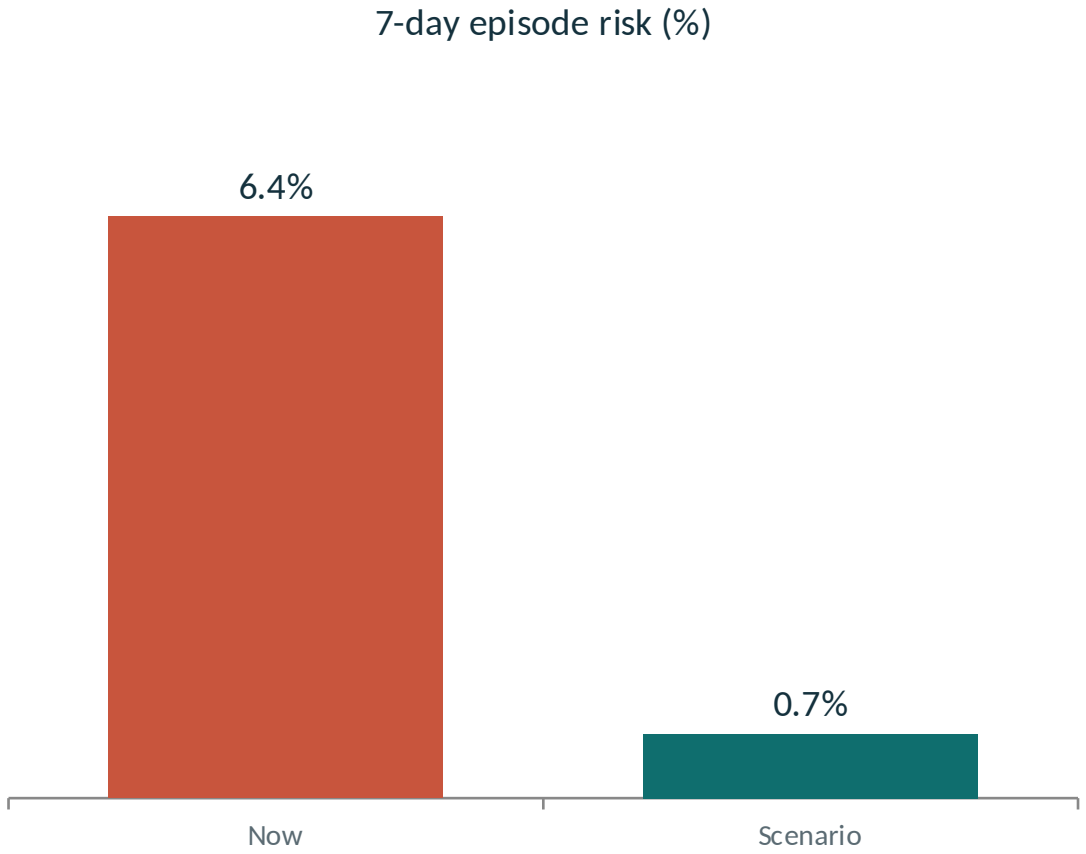
26 days of possible earlier support

Synthetic test-set patient. Risk values calculated by the trained twin model.

A healthier week moves the same student from early warning back to stable

Scenario applied to his last 7 days

- +1 hour of sleep per night
- Bedtime 45 minutes earlier
- +3,000 steps a day
- 45 fewer minutes on the phone after 11 pm
- +5 social contacts a day



Model-based sensitivity analysis on the synthetic test patient: shows how the estimate responds, not a guaranteed causal effect.

Safe by design: consent-first, human-in-the-loop, and no suicide-risk prediction

Consent-first

Explicit, revocable consent aligned with DPDP Act 2023 and ABDM consent flows

Human in the loop

The twin alerts; a counsellor or clinician decides. It never diagnoses

Explainable

Every alert ships with its reasons, measured against the person's own normal

Narrow scope

Predicts episode onset only. No suicide or self-harm prediction; crisis line Tele-MANAS 14416 in-app

Privacy

Pseudonymous IDs, counts not content from phones, aggregate-only views for institutions

Equity

Wearables skew urban and affluent: phone-only mode and bias audits on the roadmap

Next: train on real longitudinal data, then pilot with a university wellness centre

Now

PoC, real data, clinicians

Twin, dashboard, FHIR export; features validated on DEPRESJON (AUROC 0.82); 77% of 30 psychologists surveyed say it is much needed

Next 3 months

Real longitudinal data

Train the 7-day early warning on longitudinal research data such as GLOBEM; co-design alerts with clinicians

3–9 months

Campus pilot

Consented pilot with a university wellness centre; co-design alerts with counsellors

9–18 months

Scale

Health Connect / HealthKit ingestion, ABDM-linked records, multilingual nudges, phone-only mode

Honest limitations

Early-warning accuracy is from synthetic data (features checked on real actigraphy) · about half of episodes flagged early, so it supports rather than replaces human judgement · what-if is associative, not causal · one global model with personalised features, not yet per-person models.

Proactive, not reactive.

Better mental-health care starts with knowing what normal looks like for each of us, and noticing early when it changes.

Code, data, model and docs: github.com/Shrey-tech272/Digital-Twin_SHREYANSH-JOSHI

Run it: pip install -r requirements.txt → streamlit run app/streamlit_app.py

Demo video: link in README

Research prototype on synthetic data, not a diagnostic tool. If you are struggling, call Tele-MANAS 14416 (toll-free, 24×7).